

Project Support Session

HelmNet

Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Solution Approach and Submission Guidelines
- FAQs

HelmNet: Business Context

Workplace safety in hazardous environments like construction sites and industrial plants is crucial to prevent accidents and injuries. One of the most important safety measures is ensuring workers wear safety helmets, which protect against head injuries from falling objects and machinery. Non-compliance with helmet regulations increases the risk of serious injuries or fatalities, making effective monitoring essential, especially in large-scale operations where manual oversight is prone to errors and inefficiency.

To overcome these challenges, SafeGuard Corp plans to develop an automated image analysis system to detect whether workers are wearing safety helmets. This system will improve safety enforcement, ensuring compliance and reducing the risk of head injuries. By automating helmet monitoring, SafeGuard aims to enhance efficiency, scalability, and accuracy, ultimately fostering a safer work environment while minimizing human error in safety oversight.

HelmNet: Objective

As a data scientist at SafeGuard Corp, you are tasked with developing an image classification model that classifies images into one of two categories:

1. **With Helmet:** Workers wearing safety helmets.
2. **Without Helmet:** Workers not wearing safety helmets.

The ultimate objective is to prepare the model for deployment as part of an automated monitoring system. This system will enable real-time analysis of images and assist in ensuring compliance with workplace safety regulations.

HelmNet: Data Dictionary

The dataset consists of 4125 images, equally divided into two categories:

1. **With Helmet:** 3161 images showing workers wearing helmets
2. **Without Helmet:** 964 images showing workers not wearing helmets

Variations in Conditions: Images include diverse environments such as construction sites, factories, and industrial settings, with variations in lighting, angles, and worker postures to simulate real-world conditions.

Worker Activities: Workers are depicted in actions such as standing, using tools, or moving, ensuring robust model learning for various scenarios.

HelmNet: Q&A



This file is meant for personal use by uday.desiraju@gmail.com only.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

HelmNet: Evaluation Rubric

Section	Points
Data Overview	3
Exploratory Data Analysis	6
Data Pre-processing	3
Convolutional Neural Network (CNN) from Scratch	8
Transfer Learning with VGG-16 (Base)	8
Transfer Learning with VGG-16 (Base + FFNN)	8
Transfer Learning with VGG-16 (Base + FFNN + Data Augmentation)	8
Model Performance Comparison and Final Model Selection	4
Actionable Insights and recommendations	4
Presentation/Notebook - Overall Quality	8

HelmNet: Evaluation Rubric

Section 1: Data Overview

What to Cover?

Load the Dataset

Check the shape of the Data

Guiding Questions and Considerations

Did the dataset load correctly and completely?

Are there missing or inconsistent values?

HelmNet: Evaluation Rubric

Section 2: Exploratory Data Analysis

What to Cover?

Plot random images from each class and print their corresponding labels

Check for Class Imbalance

Key meaningful observations from EDA

Guiding Questions and Considerations

What do you observe from the images plotted?

What's the ratio of class imbalance?

What meaningful insights can be shared?

HelmNet: Evaluation Rubric

Section 3: Data Preprocessing

What to Cover?

Split the data into train, validation and test

Apply the normalization

Guiding Questions and Considerations

What proportions to choose for each split?

Why normalize after splitting the data?

HelmNet: Evaluation Rubric

Section 4: Convolutional Neural Network (CNN) from Scratch

What to Cover?

Define the Model

Define the configuration (loss function, optimizer, epochs, batch size)

Train the Model

Check and comment on the performance of the model

Guiding Questions and Considerations

What are the considerations for the network architecture?

What are the considerations for the configuration?

How many epochs to train for?

What data splits to use for performance checks?

HelmNet: Evaluation Rubric

Section 5: Transfer Learning with VGG-16 (Base)

What to Cover?

Define the Model

Define the configuration (loss function, optimizer, epochs, batch size)

Train the Model

Check and comment on the performance of the model

Guiding Questions and Considerations

How to do transfer learning?

What are the considerations for the configuration?

How many epochs to train for?

How does the performance change?

HelmNet: Evaluation Rubric

Section 6: Transfer Learning with VGG-16 (Base + FFNN)

What to Cover?

Define the Model

Define the configuration (loss function, optimizer, epochs, batch size)

Train the Model

Check and comment on the performance of the model

Guiding Questions and Considerations

What are the considerations for the FFNN?

What are the considerations for the configuration?

How many epochs to train for?

How does the performance change?

HelmNet: Evaluation Rubric

Section 7: Transfer Learning with VGG-16 (Base + FFNN + Data Augmentation)

What to Cover?

Define the Model

Define the configuration (loss function, optimizer, epochs, batch size)

Train the Model

Check and comment on the performance of the model

Guiding Questions and Considerations

What data augmentation techniques to use?

What are the considerations for the configuration?

How many epochs to train for?

How does the performance change?

HelmNet: Evaluation Rubric

Section 8: Model Performance Comparison and Final Model Selection

What to Cover?

Compare the performance of all the models built

Choose the best model from the ones built with proper reasoning

Check and comment on the performance of the final model on the test set

Guiding Questions and Considerations

How do the different models compare with each other?

What metric to use to decide the best model?

Is the performance on par with training performance?

HelmNet: Evaluation Rubric

Section 9: Actionable Insights & Recommendations

What to Cover?

Conclude with the key takeaways (actionable insights and recommendations) for the business

Guiding Questions and Considerations

What are the key things that we learned from the data that's useful for the business?

Logical next steps are a key inclusion

HelmNet: Evaluation Rubric

Section 10: Presentation/Notebook - Overall quality

Low Code Considerations

Clear structure, flow, and visual appeal - everything sits well in a story

Crispness - Focus on key points

All sections of the rubric included, even if in an appendix

Full Code Considerations

Clear structure and flow

Well-commented and executed code

All sections of the rubric included, even if in an appendix

HelmNet: Low-code Version

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Learner Notebook - Low Code* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

HelmNet: Low-code Version

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

HelmNet: Full-code Version

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution codes from scratch

Download the dataset and the *Learner Notebook - Full Code* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis performed

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

HelmNet: Full-code Version

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via one of the following ways:

Jupyter Notebook: *Download as HTML* from the **File** menu

Google Colab: Use [free online tools](#)

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

HelmNet: Q&A



HelmNet: FAQ

Do I need to convert images to grayscale when using VGG-16?

No, grayscale conversion is not needed. VGG-16 expects input images in RGB format (3 channels). If your dataset has grayscale images, you should convert them to RGB by duplicating the single channel into three channels to match the model's expected input shape.

How to select the best model after comparing all the models which are trained?

After training the models, the best model should be selected by comparing their validation accuracy, loss curves, and generalization ability. The chosen model is the one that performs well on the test set without overfitting, rather than just showing the highest training accuracy.

What values are recommended for data augmentation?

These values are recommended:

- Horizontal Flip: True - Object semantics remain unchanged under left-right reflection, improving generalization.
- Zoom: ≤ 0.1 - Small zoom prevents image distortion.
- Width Shift: ≤ 0.2 - Small horizontal shifts add robustness without moving the object out of frame.
- Height Shift: ≤ 0.2 - Small vertical shifts add robustness without moving the object out of frame.
- Rotation: 0 to 180° - Cameras may be installed at arbitrary angles, so wide rotation improves orientation invariance.



Power Ahead!

